



COMPUTER
ENGINEERING

QUEZON CITY

COURSE CODE/TITLE: CPE 010

SECTION: CPE21S1

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NAME: Paula Esguerra

INSTRUCTOR: Engr. Jimlord Quejado

SW NO. 4

Array-Based vs. STL Stack Implementations

1. PROCEDURE OUTPUT

ILOA

```
1  #include <iostream>
2  #include<queue>
3
4  void display(std::queue<char>);
5
6
7
8
9  int main(){
10
11      //create an object:
12      std::queue<char> myQ;
13
14
15
16
17      //use the enqueue operation
18      myQ.push('P');
19      myQ.push('A');
20      myQ.push('U');
21      std::cout << "current front is: " << myQ.front() << std::endl;
22      std::cout << "current back is: " << myQ.back() << std::endl;
23      display(myQ);
24      //use the dequeue operation
25      myQ.pop();
26      display(myQ);
27      myQ.pop();
28      display(myQ);
29      myQ.pop();
30
31      //check the queue if it is empty
32      std::cout<<"is the queue empty? " << myQ.empty() << std::endl;
33      return 0;
34  }
35
36  //note only use the member functions of the queue STL
37  void display(std::queue<char> copyQ){
38      //create a copy of the queue
39      std::queue<char> temp = copyQ;
40      //loop until empty
41      while(!temp.empty()){
42          // display the front
```

```

43         std::cout << " " << temp.front() << " ";
44         //dequeue the front
45         temp.pop();
46     }
47
48     //add a new line
49     std::cout<<std::endl;
50 };

```

OUTPUT:

```

PS C:\Users\TIPQC\Documents\CPE 010_Esguerra_CPE21S2\H0A#5\ILOA> cd "c:\
current front is: P
current back is: U
P A U
A U
U
is the queue empty? 1
PS C:\Users\TIPQC\Documents\CPE 010_Esguerra_CPE21S2\H0A#5\ILOA> 

```

ILOB

CONCLUSION

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: Gemini

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: Error assistance, and asking questions

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of



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concepts, etc.

Extent of Use: I pasted my work to check something and if there were an error, I will retype and try to understand the changed code.

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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